Personal Learning Journey – Maduwatta U.S.T.

Student	Student ID	Group	Member	Role
Maduwatta U.S.T.	IT24102088	2026-AI-33	Member 3	Regression Modelling

My assigned role is to establish regression benchmarks, compare model alternatives and hand over the selected fitted pipelines. The main lesson from this stage is that model complexity and tuning must earn their value through a fair validation comparison.

Contribution and evidence

The modelling work compares mean and median constants with Linear Regression, Ridge, Decision Tree, Random Forest and Gradient Boosting. Each model has initial settings and a training-CV search, with separate experiments excluding and including G1/G2. All candidates share the same 316 training students, 79 reserved test students and five shuffled validation folds. The notebooks now follow a consistent setup, baseline, initial-model, tuning and interpretation sequence.

A decision that shaped the project

MAE remains the selection criterion, while RMSE and R^2 explain other aspects of grade error. Random Forest was selected in both scenarios: mean CV MAE was 2.964 without prior grades and 0.938 with them. This difference reflects information available at different times; it is not evidence of a better early prediction. Accuracy, precision, recall, F1 and confusion counts additionally describe the fixed predicted-G3-below-10 rule, without replacing the original selection criterion.

Learning and critical judgement

A useful distinction is that fit trains one model, whereas cross_validate trains and evaluates separate fold copies without fitting the original estimator. GridSearchCV.fit performs the search and, with refit="MAE", fits the chosen pipeline on all training students. The selected estimator therefore needs no extra fit before handoff. Training-versus-validation gaps help diagnose overfitting. However, reusing the search folds for selected-setting scores can make those estimates optimistic; they are not independent final-test results.

Team handoff and next steps

Member 4 receives complete fitted pipelines and metadata containing the split, feature lists and fingerprints. Results remain inside the notebooks rather than being exported as many separate reports. Final test outcomes have already been inspected, so further model or threshold development requires new untouched evaluation data. My next priorities would be explaining the limits of small score differences and reviewing the support tradeoff alongside simple benchmarks, rather than assuming a more complicated model must be preferable.

Personal reflection

I expected the tuned Random Forest to win the main scenario clearly. The comparison was closer: Gradient Boosting (3.046) and Decision Tree (3.079) sit within about 0.12 of its 2.964 CV MAE, less than the forest's fold standard deviation of about 0.35. The fixed rule still selects Random Forest, but I learned to describe that margin as weak evidence and to keep the agreed MAE rule rather than switch metrics after seeing results. Linear Regression, the most interpretable option, was about 0.51 worse.

Evidence and AI use

Evidence: notebooks/03a through 03e; notebooks/03_regression_modelling.ipynb; src/model_utils.py. I used an AI assistant to suggest starting hyperparameter ranges; the final settings were chosen by training-CV MAE, not by the AI. AI also helped draft this report; its figures were checked against the notebook outputs.